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The Egyptian Exchange
البورصة المصرية

Predictive Analysis of EGX 30 Stock Trends (2021–2026)

Project 3: Data Analysis Proposal

Team:

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Project Proposal: Predictive Analysis of EGX30 Blue-Chip Stocks

1. Project Overview

This project investigates the historical trends, volatility patterns, and future price trajectory of the **EGX30** index. The benchmark index of the Egyptian Exchange (EGX) comprising the top 30 most liquid stocks. We will explore market volatility, sector correlations, and the effectiveness of technical indicators in the Egyptian market context. We are performing a time-series analysis to identify if the EGX30 follows an **Efficient Market Hypothesis** or if patterns like **Mean Reversion** exist.

2. Dataset Description

We will use the **EGX Egyptian Stocks Historical Data** dataset sourced from Kaggle. This dataset provides comprehensive daily OHLCV data for over 200 Egyptian publicly traded companies, including market leaders like Commercial International Bank (COMI), EFG Hermes (HRHO), and Talaat Moustafa Group (TMGH). We will analyze the OHLCV data (Open, High, Low, Close, Volume). Using Pandas we will create Daily Returns = $\frac{PRICE_t - PRICE_{t-1}}{PRICE_{t-1}}$ and Volatility to serve as our primary metrics for analysis.

Source: [Dataset link](#)

Time Range: January 2021 to February 2026

Format: CSV files for each stock symbol

3. Data Quality Plan

- Handle missing trading days via forward-fill
- Remove outliers using IQR method on daily return distribution
- Normalize features using MinMaxScaler before model training
- Train/test split: 80% training (2021–2025), 20% testing (2026)
- Handling Missingness: We will identify null values using `.isnull().sum()` and apply a Forward-Fill `ffill` strategy to ensure continuity in time-series data without introducing "look-ahead" bias.
- Outlier Detection: Following the **IQR** method, we will filter out extreme flash crashes or data entry errors that fall outside 1.5 x IQR of the daily returns.

4. Research Question & Hypothesis

Research Question: How do technical indicators and historical volume trends influence the short-term price movements of blue-chip stocks in the EGX30 index?

Hypotheses:

- H_0 (Null) $\mu : = 0$ (The market is a random walk, no consistent gains).
- H_1 (Alternative): $\mu \neq 0$ (There is a statistically significant trend).

- **Volatility Hypothesis:** Major macroeconomic shifts in Egypt (e.g, currency devaluations) lead to identifiable clusters of high volatility and increased trading volume in the banking and real estate sectors.
- **Predictability Hypothesis:** Technical indicators such as the Relative Strength Index (RSI) and Moving Averages (MA) can provide statistically significant signals for short-term price reversals in the EGX30.

5. Analysis Plan

Our analysis will follow these key steps:

- **Data Preprocessing:** Cleaning the dataset, handling missing values, and aligning dates across different stock symbols.
- **Correlation Heatmaps:** Using `sns.heatmap()`, we will visualize the correlation between the top 10 stocks in the EGX30 to find Sector Clusters (e.g , do Banking and Real Estate move together?).
- **Exploratory Data Analysis (EDA):** Visualizing price trends, volume spikes, and calculating daily returns for the top 5 EGX30 companies.
- **Sector Correlation:** Analyzing how different sectors (Banking, Real Estate, Telecom) move in relation to each other.
- **Technical Analysis:** Implementing RSI, MACD to identify overbought/oversold conditions.
- **Predictive Modeling:** Using a Linear Regression model to forecast the next day's closing price.

6. Expected Outcomes

- A comprehensive Python notebook demonstrating data cleaning, visualization, and modeling.
- A set of visualizations highlighting the most influential trends in the EGX over the last 5 years.
- Poster / PowerPoint. visual summary of key findings and prediction results
- A final report summarizing our findings on market predictability and sector performance.